

25 The History of Life on Earth

KEY CONCEPTS

- 25.1** Conditions on early Earth made the origin of life possible *p.* 526
- 25.2** The fossil record documents the history of life *p.* 528
- 25.3** Key events in life's history include the origins of unicellular and multicellular organisms and the colonization of land *p.* 532
- 25.4** The rise and fall of groups of organisms reflect differences in speciation and extinction rates *p.* 537
- 25.5** Major changes in body form can result from changes in the sequences and regulation of developmental genes *p.* 544
- 25.6** Evolution is not goal oriented *p.* 547

Study Tip

Draw a time line: Draw a time line like the example shown here, and mark when the following key evolutionary events occurred: “oxygen revolution,” origin of eukaryotes, first multicellular organisms, colonization of land by large eukaryotes, origin of tetrapods, and the Permian mass extinction. Briefly describe the significance of each event. You can also add more events to your time line as you study.

First unicellular organisms
(Life on Earth begins;
all early cells are prokaryotes)



Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 25
- Figure 25.10 Walkthrough: Serial Endosymbiosis
- HHMI Video: The Making of the Fittest

For Instructors to Assign (in Item Library)

- Tutorial: Visualizing the Scale of Geologic Time
- HHMI Video: The Day the Mesozoic Died

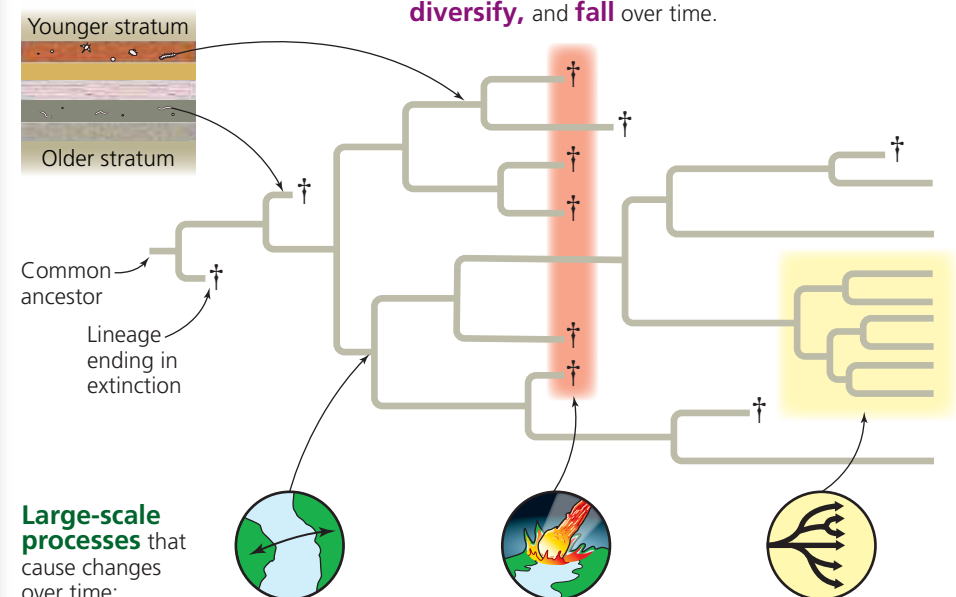


Figure 25.1 In the 1870s, researchers working in the searing heat of the Sahara Desert were startled to discover fossils of ancient whales. These fossils were spectacular not only for where they were found, but also for documenting the transition from life on land to life in the sea—just one example of the sweeping changes seen in the history of life on Earth.

How has life on Earth changed over time?

As documented by the **fossil record**...

...groups of organisms **arise, diversify, and fall** over time.



Large-scale processes that cause changes over time:

Continental drift

Land masses move over time, affecting extinction and speciation rates.

Mass extinction

Many species become extinct worldwide.

Adaptive radiation

A lineage gives rise to many new species.